

Dimension 4

$$\chi(S^3) = 0$$

Platonic cells and vertex figures

no flag bound

Schläfli Gram matrix

sign of Δ

6 regular polychora

Dimension 3

$$\chi(S^2) = 2$$

$$pF = 2E, qV = 2E$$

$$(p-2)(q-2) < 4$$

5 Platonic solids